



Spatial-temporal trend analysis of seasonal and annual rainfall (1966–2015) using innovative trend analysis method with significance test

Anurag Malik¹ · Anil Kumar¹ · Pulak Guhathakurta² · Ozgur Kisi³

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Abstract

This study investigates the spatial and temporal patterns of trends on seasonal (pre-monsoon, monsoon, post-monsoon, and winter) and annual rainfall time series data (1966–2015) at 13 stations located in the central Himalayan region of the Uttarakhand State of India. The temporal trend was analyzed using recently proposed innovative trend analysis (ITA) method with significance test. The results of the ITA method were compared with the Mann-Kendall (MK) test at 5% significance level. The spatial variation of the trends in seasonal and annual rainfall series was interpolated using the Thiessen polygon (TP) method in ArcGIS 10.2 environment. The results of comparison revealed that the trend detected by MK test (significantly positive in 3-time series and significantly negative in 6-time series) can be effectively identified using ITA (significantly positive in 19-time series and significantly negative in 32-time series). The ITA method could detect some trends that cannot be observed by the MK test. According to the spatial distribution of MK test, significantly increasing (decreasing) trends were observed in 1 (0), 1 (2), 0 (1), 1 (1), and 1 (1) polygons in pre-monsoon, monsoon, post-monsoon, winter, and annual rainfall data, while the ITA method detected significant trends in 3 (7), 6 (5), 1 (9), 4 (5), and 5 (6) polygons in the study region. The developed maps of spatial variability of rainfall trends may help the stakeholders and/or water resource managers to figure out the risk and vulnerability related to climate change in the study region.

Keywords Innovative trend analysis with significance test · Mann-Kendall · Rainfall · Thiessen polygon · Uttarakhand

Highlights

- Spatial and temporal patterns of trends on seasonal and annual rainfall time series data (1966–2015) at 13 stations located in the central Himalayan region of Uttarakhand State of India is investigated.
- The temporal trend is analyzed using a recently proposed innovative trend analysis (ITA) method with significance test.
- The results of the ITA method are compared with the Mann-Kendall (MK) method.
- The results of comparison reveal that the trend detected by the MK test (significantly positive in 3-time series and significantly negative in 6-time series) can be effectively identified using ITA (significantly positive in 19-time series and significantly negative in 32-time series).
- The ITA method could detect some trends that cannot be observed by the MK method.

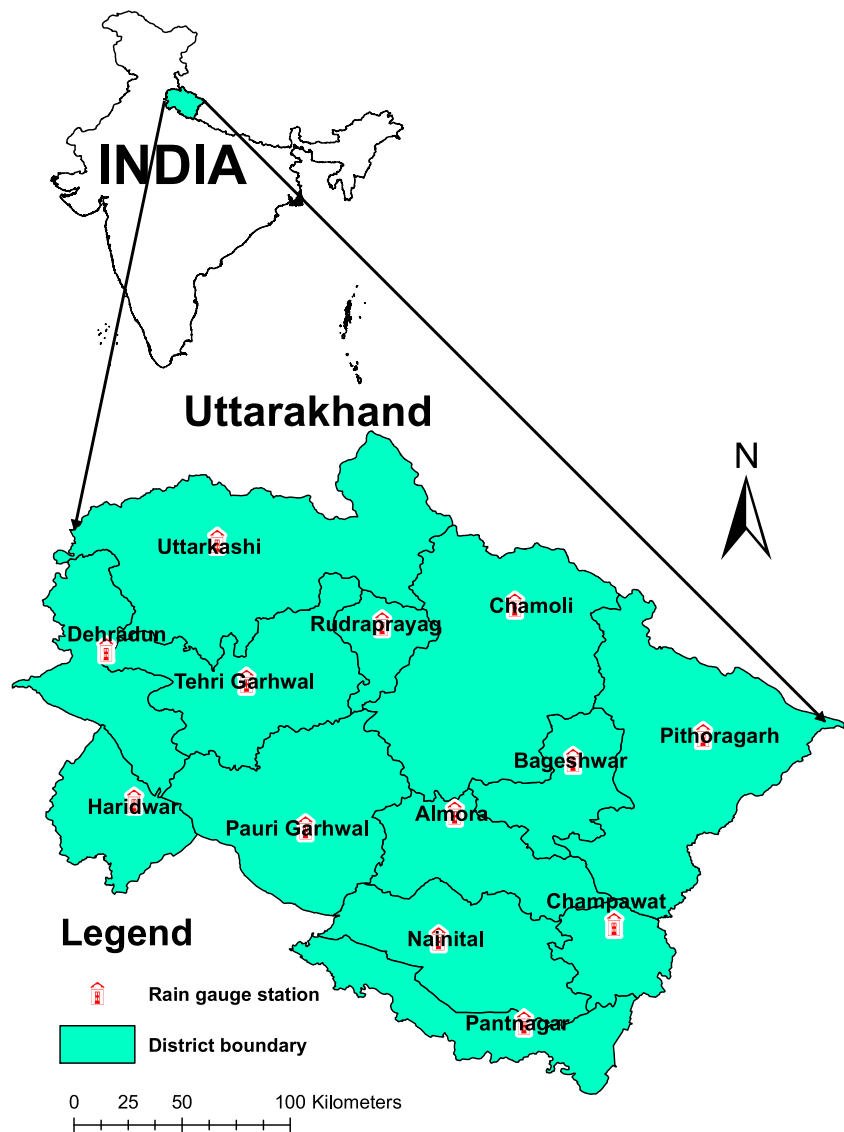
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✉ Ozgur Kisi
ozgur.kisi@iliauni.edu.ge

¹ Department of Soil and Water Conservation Engineering, College of Technology, G.B. Pant University of Agriculture and Technology, Pantnagar, Uttarakhand 263145, India

² India Meteorological Department (IMD), Pune 411005, India

³ Faculty of Natural Sciences and Engineering, Ilia State University, 0162 Tbilisi, Georgia

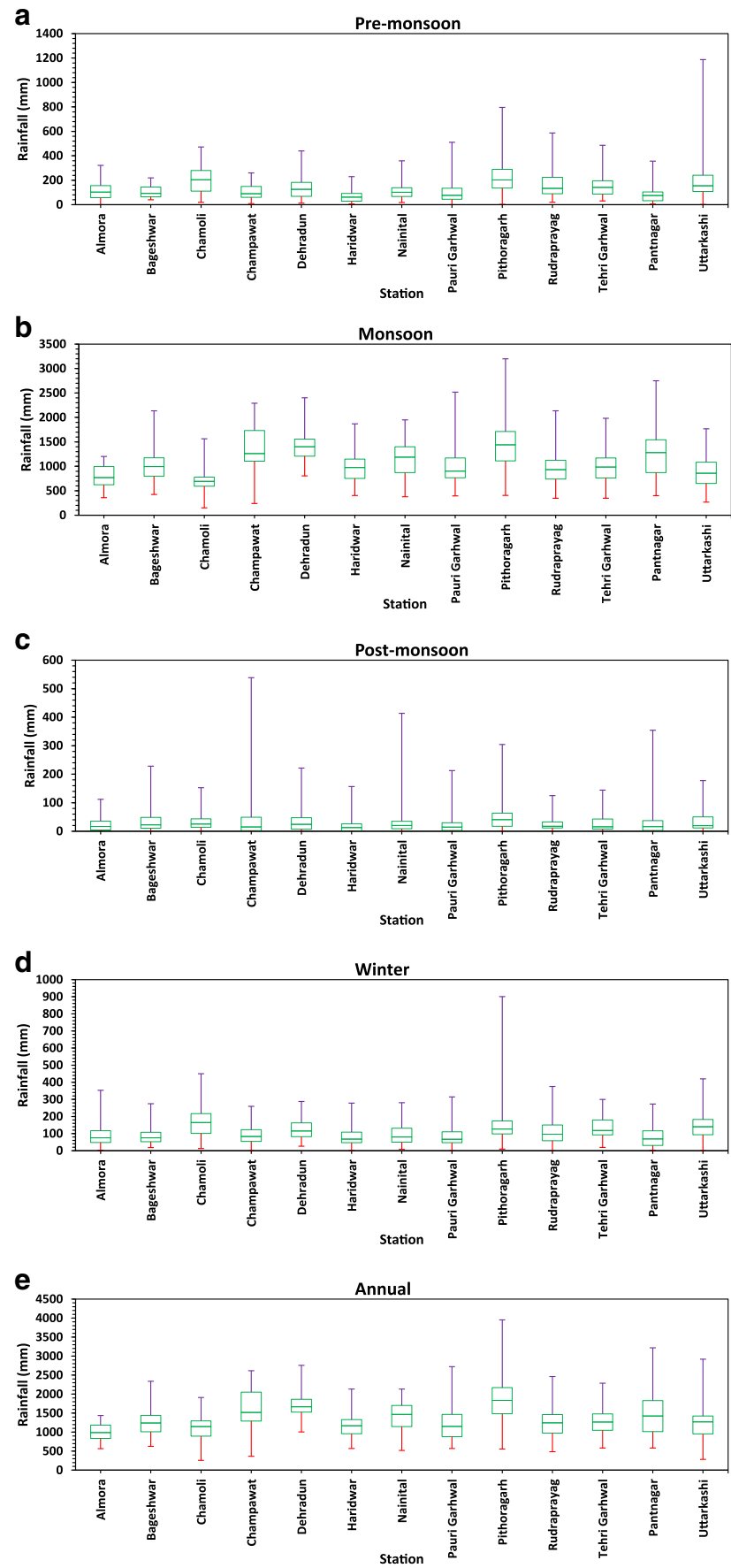
Fig. 1 Location map of the study area**Table 1** Location and data availability of rainfall at study stations

Name of station	Latitude (N)	Longitude (E)	Altitude (m)	Availability of data (year)
Almora	29° 48' 40"	79° 26' 13"	1642	1966–2015
Bageshwar	30° 05' 06"	79° 55' 30"	1004	1966–2015
Chamoli	30° 46' 30"	79° 37' 41"	2593	1966–2015
Champawat	29° 21' 54"	80° 04' 26"	1615	1966–2015
Dehradun	30° 27' 32"	77° 53' 28"	435	1966–2015
Haridwar	29° 55' 26"	78° 03' 04"	314	1966–2015
Nainital	29° 23' 20"	79° 27' 18"	2084	1966–2015
Pauri Garhwal	29° 50' 28"	78° 43' 44"	866	1966–2015
Pithoragarh	30° 11' 31"	80° 21' 54"	1514	1966–2015
Rudrapur	30° 36' 07"	79° 05' 53"	895	1966–2015
Tehri Garhwal	30° 26' 53"	78° 32' 42"	1279	1966–2015
Pantnagar	29° 00' 29"	79° 38' 02"	243	1966–2015
Uttarkashi	31° 00' 47"	78° 24' 29"	1158	1966–2015

Table 2 Descriptive statistics of seasonal and annual rainfall (1966–2015) at study stations

Statistical parameters	Almora	Bageshwar	Chamoli	Champawat	Dehradun	Haridwar	Nainital	Pauri Garhwal	Pithoragarh	Rudrapur	Tehri Garhwal	Pantnagar	Uttarkashi
Pre-monsoon													
Minimum	0.00	40.35	19.30	8.90	12.50	5.80	17.70	0.00	2.50	20.20	30.70	5.60	1.80
Maximum	321.60	219.39	472.30	259.90	440.00	229.40	359.00	510.60	796.00	586.60	486.20	356.20	1187.40
Mean	109.29	105.31	204.26	100.11	137.59	74.00	112.52	103.37	236.08	166.25	154.44	82.08	199.48
SD	61.89	49.36	113.17	56.97	86.76	60.02	73.40	95.04	154.80	114.43	88.51	63.24	172.16
Skewness	0.78	0.71	0.53	0.63	1.24	1.25	1.40	2.12	1.49	1.66	1.28	1.82	4.02
Kurtosis	1.17	-0.41	-0.06	-0.18	2.27	0.82	2.46	6.15	2.91	3.79	2.75	5.95	22.12
Monsoon													
Minimum	358.91	424.34	151.00	239.60	804.50	403.20	377.60	395.43	405.20	346.00	347.20	400.20	267.60
Maximum	1201.40	2135.23	1563.40	2291.00	2402.10	1868.70	1950.50	2517.30	3199.60	2135.90	1982.55	2750.20	1767.90
Mean	786.82	1021.17	719.62	1382.44	1431.32	982.54	1161.59	1074.77	1511.32	959.33	956.98	1283.87	875.98
SD	219.16	347.29	252.02	469.33	380.02	321.00	357.22	516.36	627.35	370.85	310.32	489.85	320.55
Skewness	0.13	1.01	0.95	0.12	0.89	0.39	-0.15	1.42	0.91	0.94	0.33	0.71	0.57
Kurtosis	-1.05	2.21	2.32	-0.34	0.71	0.02	-0.42	1.37	0.98	1.47	1.30	0.51	0.17
Post-monsoon													
Minimum	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Maximum	112.00	228.23	152.70	538.60	221.60	156.80	413.60	212.80	304.30	124.90	144.30	354.20	177.70
Mean	21.50	38.61	34.85	50.83	36.71	21.08	34.88	26.49	49.93	25.08	28.74	44.89	38.30
SD	21.53	45.56	33.27	109.15	43.59	30.46	63.79	37.38	50.25	24.38	33.99	76.06	43.44
Skewness	1.82	2.50	1.81	3.73	2.41	2.92	4.80	3.14	2.98	2.14	1.91	2.67	1.95
Kurtosis	5.19	7.34	3.65	14.18	7.08	10.07	26.51	12.66	12.85	6.25	3.53	7.27	3.51
Winter													
Minimum	3.00	18.41	13.00	0.00	26.30	1.80	7.80	0.00	9.60	0.00	19.10	2.80	0.50
Maximum	353.00	274.70	450.00	259.40	287.90	278.20	280.70	314.50	900.80	375.50	299.80	272.90	420.10
Mean	91.38	88.85	168.68	91.26	124.75	80.99	96.44	88.15	152.76	115.08	131.49	78.47	147.63
SD	69.03	50.62	96.18	55.17	61.04	52.30	60.54	69.75	129.19	74.66	65.44	59.51	79.50
Skewness	1.64	1.61	0.76	0.85	0.75	1.35	1.08	1.46	4.15	1.24	0.63	1.08	1.09
Kurtosis	3.69	3.27	0.60	0.72	0.25	2.87	0.95	1.85	23.01	2.03	0.18	1.05	2.15
Annual													
Minimum	566.20	626.38	260.00	363.20	1007.90	572.30	519.40	572.13	557.50	483.10	581.00	584.50	285.20
Maximum	1436.00	2338.76	1912.20	2615.80	2756.10	2133.70	2133.90	2720.90	3955.00	2460.60	2288.80	3218.60	2921.49
Mean	1008.09	1253.94	1127.40	1624.64	1730.37	1158.62	1405.43	1292.78	1950.09	1265.74	1271.66	1489.32	1261.39
SD	247.59	367.57	296.12	515.84	381.36	327.65	391.51	568.17	752.13	417.25	347.21	542.52	454.15
Skewness	0.00	0.79	-0.18	-0.03	0.92	0.44	-0.20	1.19	0.92	0.67	0.11	0.85	1.12
Kurtosis	-1.07	1.23	0.93	-0.44	1.38	0.48	-0.68	0.62	0.96	0.79	0.45	0.97	3.16

Fig. 2 a–e Box-whisker plot of seasonal and annual rainfall at study stations



Introduction

Climate change and anthropogenic factors adversely affect the hydro-meteorological process in a continuous manner, and their impacts appear in the forms of trends or sudden jumps (Şen 2012).

According to the Intergovernmental Panel on Climate Change based on fifth assessment (IPCC 2013), the global mean surface temperature (GMST) was increased by 0.89 °C during 1901–2012. It is sure that the minimum and maximum temperatures over land surface have been increased at global scale since 1950; but the change magnitude varied in both space and time (Zhang et al. 2008). IPCC (2013) also advised that decrease in precipitation and increase in evapotranspiration result in droughts. Rainfall is one of the dynamic parameter of the hydrological cycle, which generally varies in spatial and temporal pattern due to the increase in the concentration of atmospheric gases. The analysis of temporal and spatial variability of rainfall is important for hydrologists, agriculturalists, meteorologist, and industrialists in the aspect of sustainable utilization of water resources and control of floods and droughts, and the detailed knowledge on the spatio-temporal distribution of rainfall is crucial for accurate modeling of flood control using detention basins (Brunsell et al. 2010; Bellu et al. 2016) or surface water storage using rainwater harvesting systems (Terêncio et al. 2017, 2018). Kumar and Jain (2010) reported that the change in rainfall distribution could influence both spatial and temporal distributions of runoff, soil moisture, and groundwater reserves and could change the frequency of floods and droughts. Therefore, it is important to detect the variation of trends at spatio-temporal scales around the world (Pingale et al. 2014; Wang et al. 2013; Cui et al. 2017).

In the recent years, a number of studies have been conducted to detect the spatial-temporal trends in hydrological (streamflow) and meteorological (rainfall, evapotranspiration, temperature, humidity etc.) time series data using parametric (simple linear regression) and non-parametric (Kendall rank correlation, Spearman's rho, Mann-Kendall, modified Mann-Kendall, Thiel-Sen's slope) tests throughout the world (Tabari et al. 2011; Some'e et al. 2012; Pingale et al. 2014; Gajbhiye et al. 2015; Amirataee et al. 2016; Khatiwada et al. 2016; Sonali and Kumar 2016; Chang et al. 2017; Dhorde et al. 2017; Kumar et al. 2017; Machiwal and Jha 2017; Meshram et al. 2017; Birara et al. 2018; Bisht et al. 2018; Kabanda 2018; Machiwal et al. 2018; Dinpashoh et al. 2019; Ouatiki et al. 2019; Ray et al. 2019). Chakraborty et al. (2013) investigated the spatial and temporal variability of rainfall in Seonath basin, Chhattisgarh, India, using Mann-Kendall (MK), modified Mann-Kendall (MMK) and Spearman's rho (SR), and Thiel-Sen's slope (TSS) tests. The spatial pattern of the trend was investigated in ArcGIS environment using inverse distance weighting (IDW) method. The results indicated a downward trend in annual and seasonal rainfall. Suryavanshi et al.

(2014) analyzed spatio-temporal trends in rainfall, temperature, and potential evapotranspiration of the Betwa basin located in Central India, using the MK test. The spatial pattern of the trend was investigated in ArcGIS environment using the Thiessen polygon (TP) method. The results of analysis showed a decreasing trend in annual and seasonal precipitation, as well as maximum and minimum temperatures, and increasing trend was found in potential evapotranspiration. Pingale et al. (2014) used MK and TSS tests to detect the spatial and temporal trends of the mean (monsoon and non-monsoon seasons and annual) and extreme annual daily temperature and rainfall for the 33 urban centers of the arid and semi-arid state of Rajasthan, India. They reported that both negative and positive trends were observed in the mean and extreme events of the rainfall and temperature in the urban centers. Gajbhiye et al. (2015) analyzed the trend in monthly, seasonal, and annual rainfall data of the Sindh river basin, India, using MK and MMK tests. The magnitude of trend line was identified using TSS test. The spatial pattern of trend and its magnitude was interpolated in ArcGIS environment using a kriging technique. The significantly increasing trend was found in seasonal and annual rainfall series. Phuong et al. (2018) investigated temporal and spatial patterns of annual and seasonal rainfall data (1980–2016) at 14 weather stations located in Ho Chi Minh City, Vietnam, using linear regression, the MK, and TSS tests. The outcome showed upward trends in the seasonal and annual rainfall over the period of 37 years, but only dry season showed statistically significant trends. The results also indicated considerable increment in the dry season rainfall compared to the rainy season in most of the stations in the study region.

In the past few years, several applications of the ITA method have been found for detecting trend in hydrological and meteorological time series data (Şen 2012; Kisi and Ay 2014; Şen 2014; Ay and Kisi 2015; Kisi 2015; Dabanli et al. 2016; Cui et al. 2017; Mohorji et al. 2017; Wu and Qian 2017; Güçlü et al. 2018; Kisi et al. 2018; Wu et al. 2018; Zhou et al. 2018;

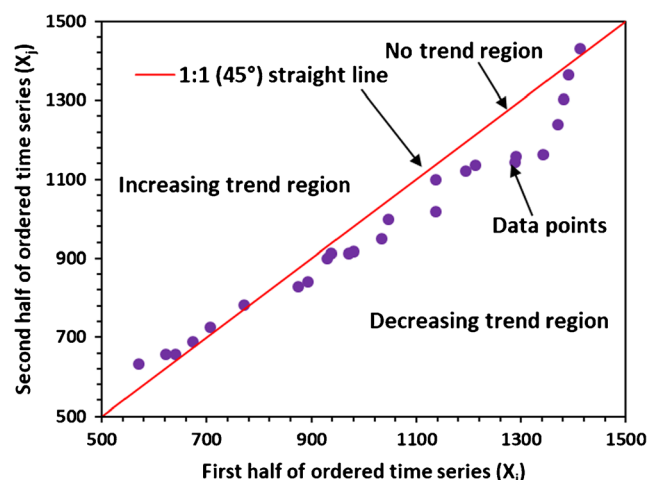


Fig. 3 The decreasing, increasing, and no trend in the ITA method

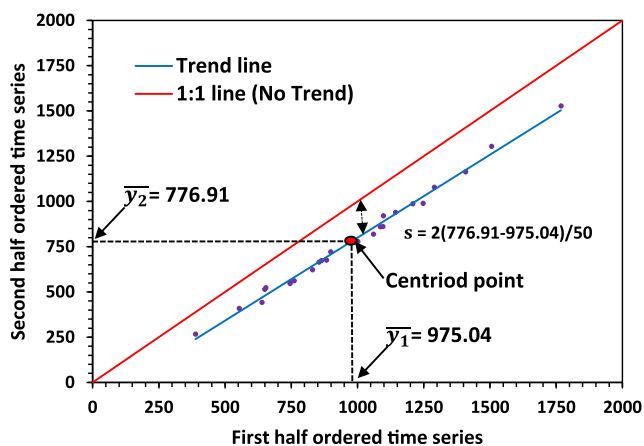
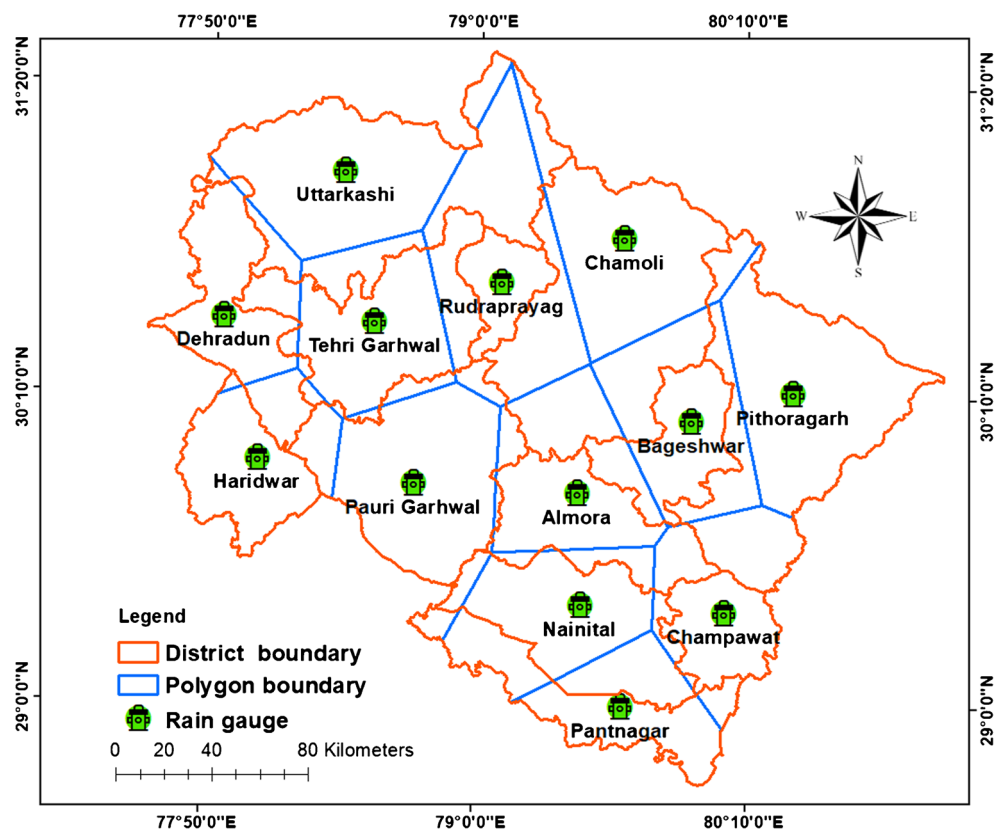


Fig. 4 The innovative trend slope

Balasmeh et al. (2019). Kisi and Ay (2014) detected a trend in water quality parameters of the Kizilirmak River, Turkey, using MK and ITA approaches at 1% and 5% levels of significance. The results of analysis revealed that the ITA method can be successfully used for trend analysis of water parameters especially in terms of evaluation of low, medium, and high values of data. Kisi (2015) identified trends in monthly pan-evaporation data of six stations (Adiyaman, Batman, Diyarbakir, Gaziantep, Kilis, and Siirt) in Turkey, using MK and ITA methods at 1% and 5% significance levels. The results of analysis revealed both increasing and decreasing trends over the study stations. He also

reported that the ITA method is not dependent on serial correlation, non-normality, and sample number, and the trends of low, medium, and high data can be easily observed by this method. Dabanli et al. (2016) applied MK and ITA methods at 5% level of significance for detecting trends in hydro-meteorological data from Ergene drainage basin, Turkey. They reported that the ITA method yields trend categorizations (very low, low, medium, high, and very high), which helps in future flood (very high) and drought (very low) studies. Cui et al. (2017) analyzed trends in annual and seasonal air temperature and rainfall in the Yangtze River Basin, China, using linear regression (LR), MK, Sen's slope estimator (SSE), and ITA methods at 10%, 5%, and 1% levels of significance. They found that the seasonal and annual maximum, minimum, and mean temperatures increase significantly. The annual precipitation increases significantly while the seasonal precipitation significantly decreases. All the studies above used the first version of the ITA method. Only one study conducted by Zhou et al. (2018) investigated trends in annual and seasonal solar radiation at 48 stations in five different climatic zones across China during 1962–2015, using LR, MK, and ITA approaches with significance test recently developed by Sen (2017) at 1% and 5% significance levels. The results of analysis indicated increasing and decreasing trends in annual and seasonal solar radiation over study stations. They also reported that the many significant trends that cannot be effectively detected by LR and MK tests can be identified

Fig. 5 Thiessen polygon map of the study area



using the new ITA method. Caloiero et al. (2018) investigated temporal trend in monthly, seasonal, and annual rainfall data of 50 years in southern Italy, using MK and ITA methods at 5% significance level. They found trends in seasonal and annual rainfall values using the ITA method, while the MK test indicated a negative trend in winter and autumn rainfall data over the study region.

Keeping the above reviews in mind, this study was conducted with the following specific objectives: (i) to analyze temporal trend in seasonal and annual rainfall time series data using the MK and new ITA method with significance test; (ii) to compare the results of MK method with the new ITA; and (ii) to analyze the spatial pattern of trends in seasonal and annual rainfall using the Thiessen polygon method in

Table 3 Station with area under the district, Thiessen polygon, and percent area of contributing districts in each polygon

Name of district/polygon	District area (km ²)	Polygon area (km ²)	Contributing area of districts in the polygon		
			District	Area (km ²)	(%)
Almora	3101.20	4046.47	Almora	2128.44	52.60
			Nainital	9.17	0.23
			Pauri Garhwal	274.60	6.79
			Chamoli	1131.92	27.97
Bageshwar	2280.13	4064.40	Bageshwar	502.34	12.41
			Bageshwar	1777.48	43.73
			Almora	4.59	0.11
			Chamoli	1330.08	32.73
Chamoli	7926.17	4654.13	Pithoragarh	952.24	23.43
			Chamoli	3950.58	84.88
			Pithoragarh	323.56	6.95
			Uttarkashi	379.98	8.16
Champawat	1683.54	3747.18	Champawat	1682.66	44.90
			Udham Singh Nagar	220.03	5.87
			Nainital	214.81	5.73
			Almora	499.97	13.34
Dehradun	3071.00	3295.78	Bageshwar	0.31	0.01
			Pithoragarh	1129.41	30.14
			Dehradun	2410.70	73.15
			Haridwar	40.45	1.23
Haridwar	2388.81	3475.18	Tehri Garhwal	455.29	13.81
			Uttarkashi	389.35	11.81
			Haridwar	2348.37	67.58
			Pauri Garhwal	488.91	14.07
Nainital	4062.88	3888.39	Tehri Garhwal	67.79	1.95
			Dehradun	570.11	16.41
			Nainital	2626.44	67.55
			Udham Singh Nagar	813.29	20.92
Pauri Garhwal	5457.66	4721.67	Almora	448.65	11.54
			Pauri Garhwal	4281.49	90.68
			Almora	19.55	0.41
			Nainital	206.67	4.38
Pithoragarh	7324.11	4918.90	Udham Singh Nagar	163.58	3.46
			Tehri Garhwal	50.38	1.07
			Pithoragarh	4918.90	100
			Rudraprayag	1214.08	21.46
Rudraprayag	1214.08	5658.50	Pauri Garhwal	207.63	3.67
			Tehri Garhwal	899.71	15.90
			Uttarkashi	1823.49	32.23
			Chamoli	1513.59	26.75
Tehri Garhwal	4391.38	3838.52	Tehri Garhwal	2918.20	76.02
			Dehradun	44.25	1.15
			Pauri Garhwal	205.02	5.34
			Uttarkashi	671.04	17.48
Pantnagar	2861.64	2671.40	Udham Singh Nagar	1664.74	62.32
			Nainital	1005.79	37.65
			Champawat	0.88	0.03
			Uttarkashi	4786.02	99.05
Uttarkashi	8049.87	4831.96	Dehradun	45.94	0.95

ArcGIS 10.2 environment. It is worthy to note that this is the first study that applies the new ITA with significance test to rainfall data.

Materials and methods

Description of study area

This study was conducted on 13 districts of the Uttarakhand State located in the central Himalayan region of India, with a total geographical area of 53,812.47 km² as shown in Fig. 1. The Uttarakhand state lies between 77° 34' E and 81° 03' E longitudes and 28° 43' N and 31° 28' N latitudes with altitudes varying from 175 to 7409 m above mean sea level. The state of Uttarakhand shares its boundaries with Himachal Pradesh in the northwest, Uttar Pradesh in south, and with Nepal and China in the southeast and northeast, respectively. The state is grouped into 13 districts and split into two administrative divisions such as the (i) Garhwal region including seven districts namely Chamoli, Dehradun, Haridwar, Pauri Garhwal, Rudraprayag, Tehri Garhwal, and Uttarkashi and (ii) Kumaon region including six districts namely Almora, Bageshwar, Champawat, Nainital, Pithoragarh, and Udham Singh Nagar (Fig. 1). Uttarakhand is split into four physiographic zones viz.: (i) Terai (finer alluvium deposits) and Bhabar (a belt of pebbles) region between the elevations of 175 to 600 m; (ii) Shivalik region between the elevations of 600 to 1200 m; (iii) Lesser Himalaya region between the elevations of 1200 and 3000 m; and (iv) Great Himalaya region between the elevations of 3000 and 7000 m above mean sea level. The state has a temperate climate, except in the plains

where the climate is tropical. The temperature in the state ranges from sub-zero to 43 °C. The annual rainfall varies between 260 and 3955 mm, of which most part (60 to 85% of annual total) is received in the monsoon season (June to September) alone. The details of latitude, longitude, altitude, and data period are given in Table 1.

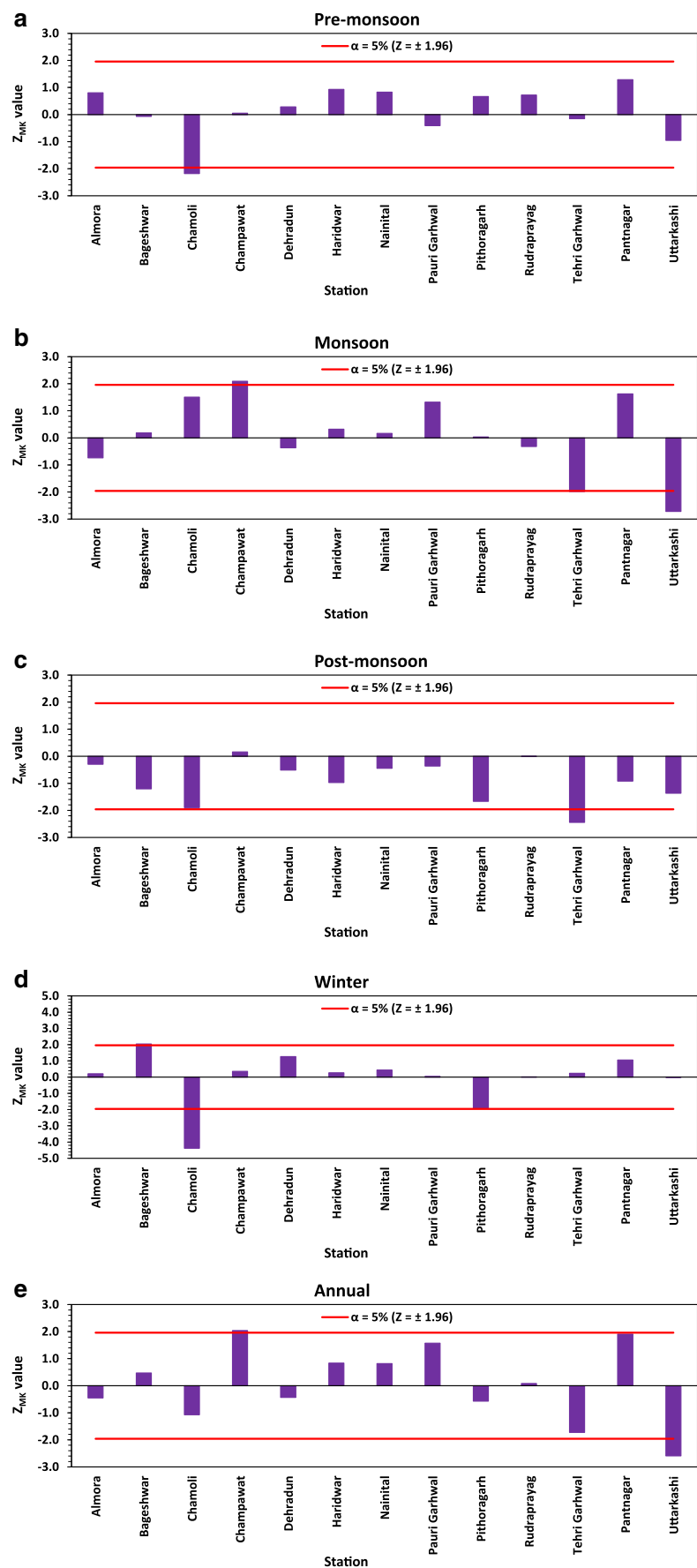
The monthly rainfall (mm) time series data of 13 districts were collected from Indian Meteorological Department, Pune. According to the Indian Meteorological Department, four prominent seasons namely (i) winter (December–February), (ii) pre-monsoon (March–May), (iii) monsoon (June–September), and (iv) post-monsoon (October–November) are dominant in India. In this study, the analysis of trend was carried on seasonal (pre-monsoon, monsoon, post-monsoon, and winter) and annual time scales. The statistical analysis results (minimum, maximum, mean, standard deviation, skewness, and kurtosis) of monthly, seasonal, and annual rainfall data are given in Table 2. It is apparent from the table that rainfall generally has positive skewness. The data have higher skewness in the post-monsoon season compared to other seasons. In some cases, rainfall has highly skewed distribution (e.g., Uttarkashi in pre-monsoon, Nainital in post-monsoon, Pithoragarh in winter). The graphical representation of seasonal (pre-monsoon, monsoon, post-monsoon, and winter) and annual rainfall time series data in the form of box-whisker plots are illustrated in Fig. 2a–e), respectively. They provide information about the minimum, first quartile, median, third quartile, and maximum values of meteorological data. It is clear from Fig. 2 that monsoon season has much higher rainfall compared to other seasons. The plots of monsoon and annual rainfall are similar to each other and this indicates the effect of monsoon season data to annual ones.

Table 4 The results of MK test on seasonal and annual rainfall data at study stations

Station	MK test statistic (Z_{MK})				
	Pre-monsoon	Monsoon	Post-monsoon	Winter	Annual
Almora	0.803	−0.736	−0.293	0.201	−0.452
Bageshwar	−0.067	0.184	−1.205	2.041*	0.468
Chamoli	−2.183*	1.506	−1.899	−4.383*	−1.071
Champawat	0.050	2.091*	0.159	0.351	2.041*
Dehradun	0.284	−0.368	−0.510	1.263	−0.435
Haridwar	0.929	0.318	−0.972	0.268	0.836
Nainital	0.828	0.167	−0.443	0.435	0.820
Pauri Garhwal	−0.410	1.322	−0.360	0.050	1.573
Pithoragarh	0.669	0.033	−1.665	−1.916	−0.569
Rudraprayag	0.719	−0.318	−0.008	0.017	0.084
Tehri Garhwal	−0.151	−1.983*	−2.443*	0.234	−1.723
Pantnagar	1.288	1.623	−0.915	1.054	1.907
Uttarkashi	−0.954	−2.719*	−1.364	−0.017	−2.593*

*Statistically significant at $\alpha = 5\%$ (critical $Z = \pm 1.96$)

Fig. 6 a–e Variation of calculated Z_{MK} values of seasonal and annual rainfall at 5% s.l. using the MK test at study stations



Mann-Kendall test

The Mann-Kendall (MK) test (Mann 1945; Kendall 1975) is a rank based non-parametric test to detect significant trend in meteorological and hydrological time series data (Tabari et al. 2011; Some'e et al. 2012; Suryavanshi et al. 2014; Pingale et al. 2014; Gajbhiye et al. 2015; Kumar et al. 2017). The MK test is suitable for the non-autocorrelated data and does not require the assumption of normality or homogeneity of variance. This test compares median rather than mean and, as a result, if the data have one or two outliers, their influence is neglected. The Mann-Kendall test statistic (S) is defined as:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sign}(x_j - x_i) \quad (1)$$

where, n is the number of observations, x_i is the rank for i th observations ($i = 1, 2, \dots, n-1$), x_j is the rank for j th observations ($j = i+1, 2, \dots, n$), and sign function is computed as:

$$\text{sign}(x_j - x_i) = \begin{cases} 1; & \text{if } (x_j - x_i) > 0 \\ 0; & \text{if } (x_j - x_i) = 0 \\ -1; & \text{if } (x_j - x_i) < 0 \end{cases} \quad (2)$$

The test statistic S is assumed for the series where sample size $n > 10$ is asymptotically normally distributed with mean $E(S)$ and variance $\text{Var}(S)$ as:

$E(S) = 0$, and

$$\text{Var}(S) = \frac{n(n-1)(2n+5)}{18} \quad (3)$$

If there is a possibility of tie in the value of x , the variance is computed as:

$$\text{Var}(S) = \frac{n(n-1)(2n+5) - \sum_{i=1}^P t_i(t_i-1)(2t_i+5)}{18} \quad (4)$$

where P is the number of tied groups, $(\)$ indicates the summation over all tied groups, and t_i is the number of data values in the i th tied group ($i = 1, 2, 3, \dots, n$). After the calculation of $\text{Var}(S)$ of time series data, the standardized Z_{MK} value is calculated using the equation:

$$Z_{\text{MK}} = \begin{cases} \frac{S-1}{\sqrt{\text{Var}(S)}}; & \text{if } S > 0 \\ 0; & \text{if } S = 0 \\ \frac{S+1}{\sqrt{\text{Var}(S)}}; & \text{if } S < 0 \end{cases} \quad (5)$$

The standardized MK test statistic (Z_{MK}) follows the standard normal distribution (Z) with mean zero ($\mu = 0$) and variance one ($\sigma^2 = 1$). The null hypothesis (H_0) is such that there is no trend in the rainfall time series, and the alternate hypothesis (H_1) is such that there is a trend in the rainfall time series. Thus, in a two-sided condition, the H_0 and H_1 are tested at one significance levels (α), i.e., $\alpha = 5\%$ with $Z = \pm 1.96$. If $\pm Z_{\text{MK}} > \pm Z_{\alpha/2}$, then H_0 is rejected and H_1 is accepted. A positive (negative) value of Z_{MK} indicates an increasing (decreasing) trend in the time series.

Innovative trend analysis method

Şen (2012) proposed the concept of innovative trend analysis (ITA) and later used by Kisi and Ay (2014), Şen (2014), Ay and Kisi (2015), Kisi (2015), Dabanli et al. (2016), Cui et al. (2017), Mohorji et al. (2017), Wu and Qian (2017), Güçlü et al. (2018), Kisi et al. (2018), Wu et al. (2018), and Zhou et al. (2018)). In this method, a time series of data is divided into two equal halves from the first date to the end date. Both the sub-series are sorted in ascending order; the first half series (X_i ; $i = 1, 2, \dots, n/2$) is located on horizontal axis (X -axis) and the second half series (X_j ; $j = n/2 + 1, n/2 + 2, \dots, n$) is located on the vertical

Table 5 The results of innovative trend analysis on pre-monsoon rainfall at study stations

Station	Slope (s)	σ	$\rho_{y_1 y_2}$	σ_s	Upper CL	Lower CL
Almora	0.21	61.89	0.938	0.12	0.24	-0.24
Bageshwar	-0.40*	49.36	0.989	0.04	0.07	-0.07
Chamoli	-3.12*	113.17	0.947	0.21	0.41	-0.41
Champawat	-0.31*	56.97	0.979	0.07	0.13	-0.13
Dehradun	-0.59*	86.76	0.940	0.17	0.33	-0.33
Haridwar	0.28*	60.02	0.986	0.06	0.11	-0.11
Nainital	-0.33*	73.40	0.979	0.08	0.16	-0.16
Pauri Garhwal	-0.03	95.04	0.932	0.20	0.39	-0.39
Pithoragarh	2.14*	154.80	0.971	0.21	0.41	-0.41
Rudraprayag	-0.23	114.43	0.974	0.15	0.29	-0.29
Tehri Garhwal	-0.18	88.51	0.910	0.21	0.42	-0.42
Pantnagar	0.60*	63.24	0.878	0.18	0.35	-0.35
Uttarkashi	-3.18*	172.16	0.930	0.73	1.43	-1.43

*Statistically significant at $\alpha = 5\%$ (critical $Z = \pm 1.96$)

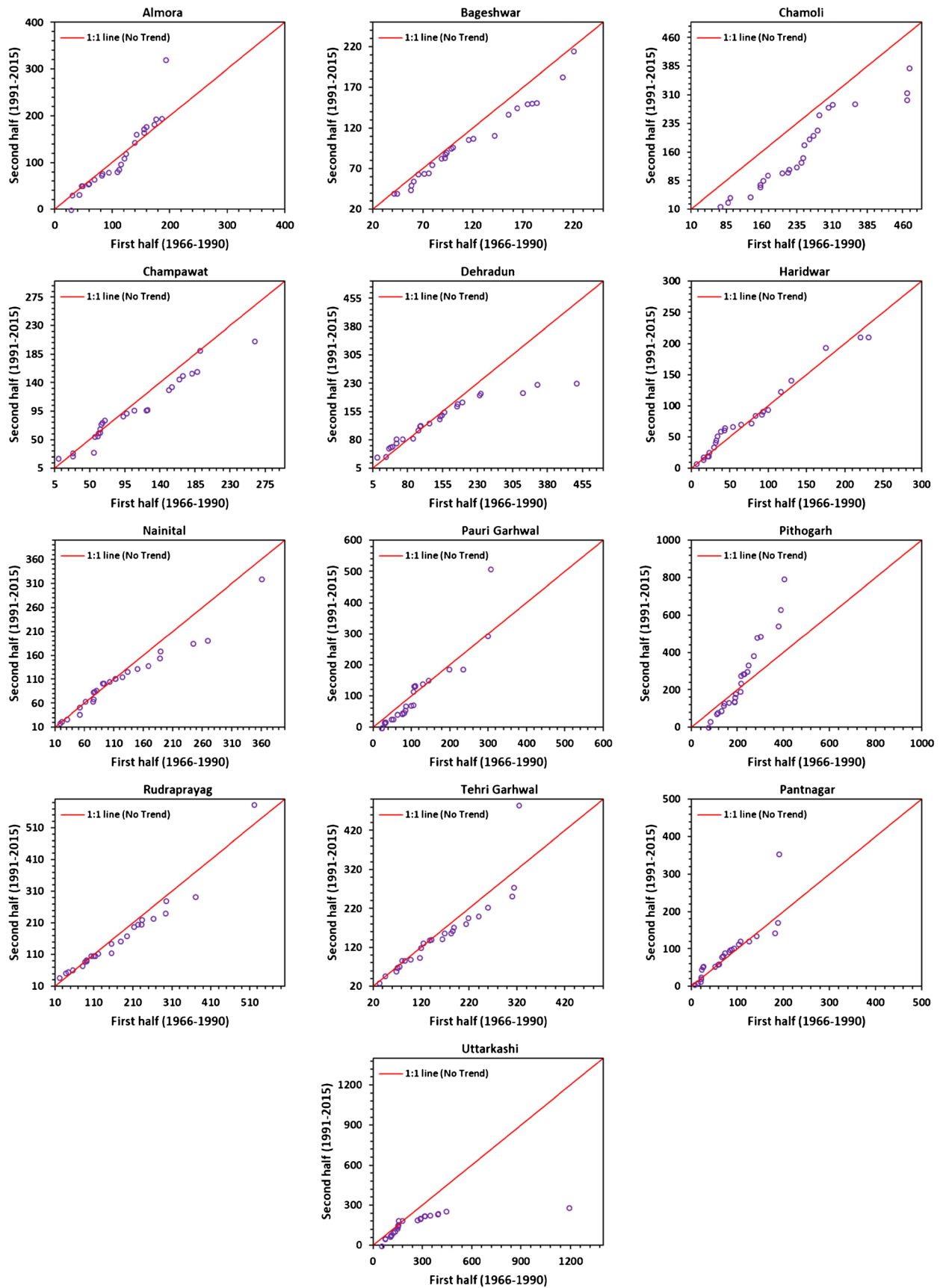


Fig. 7 The results of ITA for pre-monsoon rainfall at study stations

axis (Y-axis) of the Cartesian coordinate system (Fig. 3). The range of both the axis should be equal. The 1:1 (45°) line divides the diagram (Fig. 3) into two equal triangles, where the area of the upper and lower triangle indicate increasing and decreasing trends into the time series data (Şen 2012, 2014). If the points in scatter plot accumulate on 1:1 (45°) line or close to this line, this indicates no significant trend in the time series. If the points in scatter plot fall above and below the 1:1 (45°) line, this indicates the non-monotonic trends (may be positive or negative) in the time series. Then, it is possible to categorize the variation domain of each half series in several groups (low, medium, and high). Such plot represents first visual inspection of the type of trends. The range of each group can be determined either by an expert view or quantitatively by splitting the domain of variation into some sub-classes (Dabanli et al. 2016). The closer the scatter points to the 1:1 (45°) line, the weaker the trend slope (Şen 2012).

Şen (2017) tested the ITA method at 1%, 5%, and 10% significance levels for a better understanding of existing trend nature into the time series of data. The trend slope is computed using the following expression (Şen 2017):

$$s = \frac{2(\bar{y}_2 - \bar{y}_1)}{n} \quad (6)$$

where s is the slope of trend line; $\bar{y}_2$ and $\bar{y}_1$ are the arithmetic averages of the first and second halves of the dependent variable (y , sequence); and n is the number of data year. Figure 4 is the graphical presentation for understanding the concept of trend slope.

After the calculation of slope of the trend line, the standard deviation of sampling slope is calculated as:

$$\sigma_s = \frac{2\sqrt{2}}{n\sqrt{n}} \sigma \sqrt{1 - \rho_{\bar{y}_1 \bar{y}_2}} \quad (7)$$

where σ_s is the standard deviation of sampling slope; σ is the standard deviation of whole data series; and $\rho_{\bar{y}_1 \bar{y}_2}$ is the cross-correlation coefficient between the ascendingly sorted two halves' arithmetic averages.

Finally, the confidence limits (upper and lower) of the trend slope at α level of significance are expressed using the following expression:

$$CL_{1-\alpha} = 0 \pm s_{crit} \sigma_s \quad (8)$$

where CL is the upper and lower confidence limits at α level of significance; σ_s is the standard deviation of sampling slope; and s is the slope value. The null hypothesis (H_0) is such that there is no trend in the time series, and alternate hypothesis (H_1) is such that there is a trend in the time series. Thus, in a two-sided condition, the H_0 and H_1 are tested at $\alpha = 5\%$ with $Z = \pm 1.96$. If $\pm s > \pm CL_{1-\alpha}$, then H_0 is rejected and H_1 is accepted. A positive (negative) value of s indicates an increasing (decreasing) trend in the time series (Şen 2017; Zhou et al. 2018).

Thiessen polygon

Thiessen (1911) proposed the concept of Thiessen polygon (TP) method for spatial interpolation which is based on proximal mapping, i.e., the nearest distance neighbor (Basistha et al. 2008; Suryavanshi et al. 2014; Hadi and Tombul 2018). In this study, 13 rain gauges are installed over the study region for rainfall recording, and the spatial coverage of each station was determined using the TP method in ArcGIS 10.2 environment (Fig. 5). Table 3 gives the detailed information about the areas under each station, district, Thiessen polygon, and percent contribution of different districts within each polygon. These 13 polygons were used for spatial mapping of the trends detected using MK and ITA methods over the study area.

Table 6 The results of innovative trend analysis on monsoon rainfall at study stations

Station	Slope (s)	σ	$\rho_{\bar{y}_1 \bar{y}_2}$	σ_s	Upper CL	Lower CL
Almora	-1.62*	219.16	0.980	0.25	0.48	-0.48
Bageshwar	0.39	347.29	0.936	0.70	1.38	-1.38
Chamoli	3.61*	252.02	0.933	0.52	1.03	-1.03
Champawat	8.31*	469.33	0.932	0.98	1.91	-1.91
Dehradun	-2.12*	380.02	0.971	0.51	1.01	-1.01
Haridwar	1.65*	321.00	0.972	0.43	0.84	-0.84
Nainital	-0.86	357.22	0.938	0.71	1.39	-1.39
Pauri Garhwal	14.59*	516.36	0.937	1.04	2.04	-2.04
Pithoragarh	6.57*	627.35	0.940	1.22	2.40	-2.40
Rudraprayag	-3.67*	370.85	0.972	0.50	0.97	-0.97
Tehri Garhwal	-4.40*	310.32	0.935	0.63	1.24	-1.24
Pantnagar	7.05*	489.85	0.929	1.05	2.05	-2.05
Uttarkashi	-7.93*	320.55	0.997	0.15	0.30	-0.30

*Statistically significant at $\alpha = 5\%$ (critical $Z = \pm 1.96$)

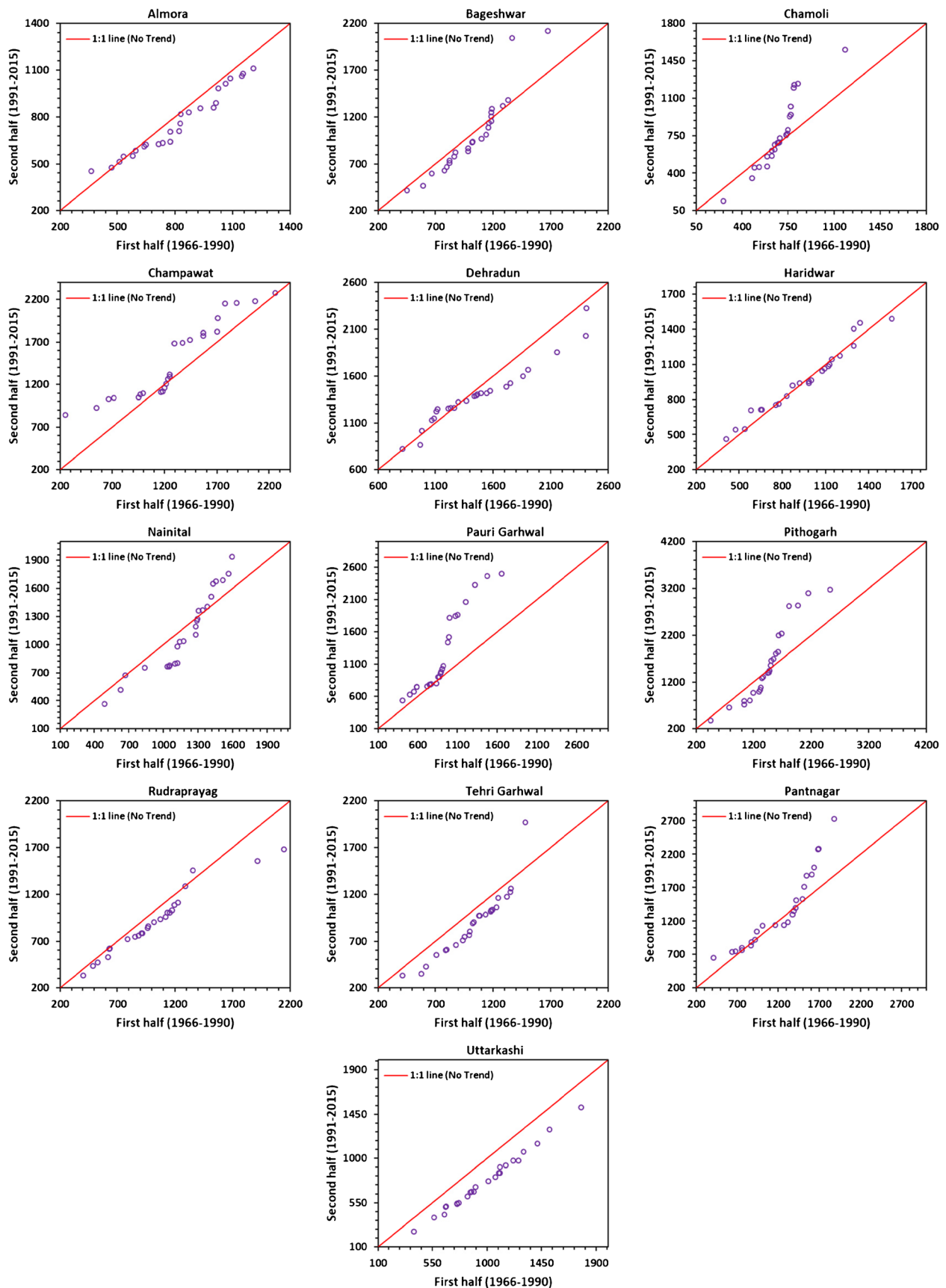


Fig. 8 The results of ITA for monsoon rainfall at study stations

Results and discussion

The temporal trend analysis of seasonal (pre-monsoon, monsoon, post-monsoon, and winter) and annual rainfall time series was performed using Mann-Kendall (MK) and innovative trend analysis (ITA) methods at 5% level of significance for all the stations under consideration. The spatial distribution of seasonal and annual trends values of MK and ITA were indicated using the Thiessen polygon areal demarcation method. The trend analysis in seasonal (pre-monsoon, monsoon, post-monsoon, and winter) and annual rainfall series is described in the following subsequent sections.

Temporal variation of trends in pre-monsoon rainfall data series

The MK test was applied to the pre-monsoon rainfall data series and the values of the test statistic (Z_{MK}) for all the 13 stations are summarized in Table 4 along with their graphical representation in Fig. 6a. Figure 6a shows the calculated Z_{MK} values for pre-monsoon rainfall series and tested at 5% significance level. If the vertical bars cross these set limits, there is a significant positive or negative trend in the rainfall data, otherwise no trend. It is observed from Table 4 that the positive values are more dominant but are not statistically significant at 5% level. The significant negative trend was found for Chamoli station (-2.183) at 5% level in pre-monsoon season.

After the MK test, the ITA was applied to the pre-monsoon season rainfall data and the values of test for all the 13 stations are summarized in Table 5. It can be observed from the Table 5 that the negative values of slope (s) are more dominant, and most of them are statistically significant at 5% significance

level. The significant negative trend was found for Bageshwar (-0.40), Chamoli (-3.12), Champawat (-0.31), Dehradun (-0.59), Nainital (-0.33), and Uttarkashi (-3.18) stations while a significant positive trend was found for Haridwar (0.28), Pithoragarh (2.14), and Pantnagar (0.60) stations at 5% level in pre-monsoon season. The graphical results of the ITA method for all the 13 stations during pre-monsoon season (1966–2015) are illustrated in Fig. 7.

Temporal variation of trends in monsoon rainfall data series

The rainfall trend in monsoon season was detected using the MK test and the values of test statistic are provided in Table 4 along with their graphical representation in Fig. 6b. Figure 6b demonstrates the calculated Z_{MK} values for monsoon rainfall series and tested at 5% significance level. The significant positive trend was found for Champawat (2.091) station while the significant negative trend was found for Tehri Garhwal (-1.983) and Uttarkashi (-2.719) stations at 5% level in monsoon season.

After the application of the MK test, the ITA was applied to the monsoon season rainfall data and the values of the test are given in Table 6. The significant negative trend was found for Almora (-1.62), Dehradun (-2.12), Rudraprayag (-3.67), Tehri Garhwal (-4.40), and Uttarkashi (-7.93) stations while a significant positive trend was found for Chamoli (3.61), Champawat (8.31), Haridwar (1.65), Pauri Garhwal (14.59), Pithoragarh (6.57), and Pantnagar (7.05) stations at 5% level in monsoon season. The graphical results of the ITA method for all the 13 stations during monsoon season (1966–2015) are illustrated in Fig. 8. As observed from Fig. 8, for Bageshwar and Nainital stations, no significant trend was detected at 5% level in monsoon season rainfall.

Table 7 The results of innovative trend analysis on post-monsoon rainfall at study stations

Station	Slope (s)	σ	$\rho_{\overline{y_1} \overline{y_2}}$	σ_s	Upper CL	Lower CL
Almora	-0.33^*	21.53	0.901	0.05	0.11	-0.11
Bageshwar	-0.36^*	45.56	0.937	0.09	0.18	-0.18
Chamoli	-0.92^*	33.27	0.949	0.06	0.12	-0.12
Champawat	-0.03	109.15	0.946	0.20	0.40	-0.40
Dehradun	-0.62^*	43.59	0.965	0.07	0.13	-0.13
Haridwar	-0.24^*	30.46	0.963	0.05	0.09	-0.09
Nainital	-0.48^*	63.79	0.940	0.12	0.24	-0.24
Pauri Garhwal	0.25^*	37.38	0.987	0.03	0.07	-0.07
Pithoragarh	-0.96^*	50.25	0.968	0.07	0.14	-0.14
Rudraprayag	-0.002	24.38	0.975	0.03	0.06	-0.06
Tehri Garhwal	-0.78^*	33.99	0.897	0.09	0.17	-0.17
Pantnagar	0.16	76.06	0.925	0.17	0.33	-0.33
Uttarkashi	-0.99^*	43.44	0.834	0.14	0.28	-0.28

*Statistically significant at $\alpha = 5\%$ (critical $Z = \pm 1.96$)

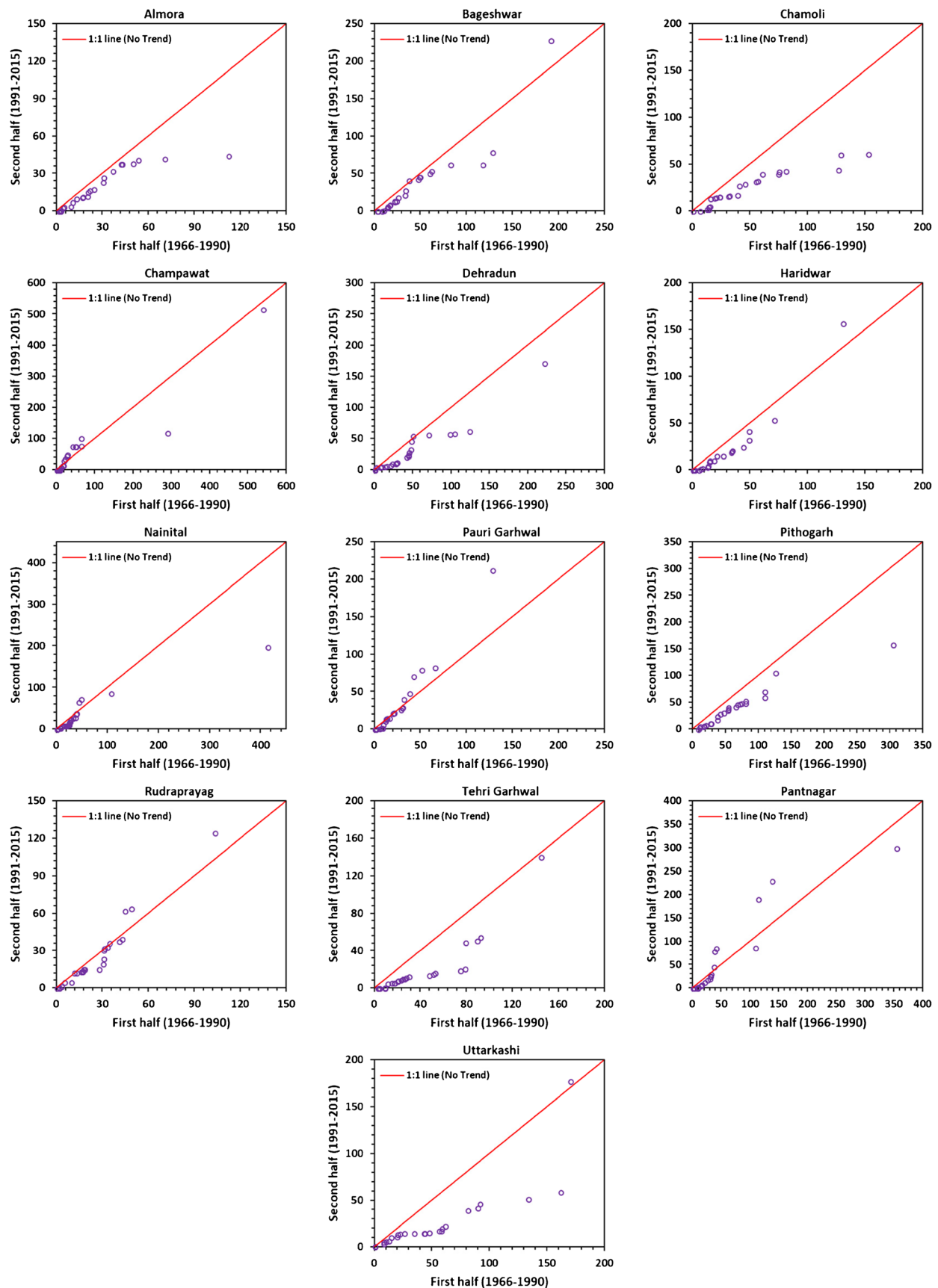


Fig. 9 The results of ITA for post-monsoon rainfall at study stations

Temporal variation of trends in post-monsoon rainfall data series

The MK test statistics obtained from post-monsoon rainfall data series are given in Table 4 along with their graphical representation in Fig. 6c. Figure 6c depicts the calculated Z_{MK} values for post-monsoon rainfall series and tested at 5% significance level. It is observed from the table that the negative values are more dominant and most of them are statistically not significant at 5% significance level. The significant negative trend was found for Tehri Garhwal (−2.443) station at 5% level in post-monsoon season.

The test values of ITA for all the 13 stations are summarized in Table 7. It is observed from the table that the negative values of slope (s) are more dominant, and most of them are statistically significant at 5% significance level. The significant negative trend was found for Almora (−0.33), Bageshwar (−0.36), Chamoli (−0.92), Dehradun (−0.62), Haridwar (−0.24), Nainital (−0.48), Pithoragarh (−0.96), Tehri Garhwal (−0.78), and Uttarkashi (−0.99) stations while a significant positive trend was found for Pauri Garhwal (0.25) station at 5% level in post-monsoon season. The graphical results of the ITA method for all the 13 stations during pre-monsoon season (1966–2015) are illustrated in Fig. 9.

Temporal variation of trends in winter rainfall data series

The MK test statistics obtained from winter season rainfall time series are summarized in Table 4 along with their graphical representation in Fig. 6d. Figure 6d illustrates the calculated Z_{MK} values for winter rainfall series and tested at 5%

significance level. As observed from Table 4 that the positive values are more dominant, some of them are statistically significant at the 5% significance level. The significant positive trend was found for Bageshwar (2.041) station while the significant negative trend was found for Chamoli (−4.383) station at 5% level in the winter season. The value of Pithoragarh (−1.916) is also very close to significance limit (−1.96).

The ITA values are provided in Table 8 and graphically illustrated in Fig. 10. As observed from the table, the positive values of slope (s) are more dominant, and most of them are statistically significant at 5% significance level. The significant negative trend was found for Chamoli (−4.05), Champawat (−0.27), Pithoragarh (−2.00), Rudraprayag (−1.01), and Tehri Garhwal (−0.41) stations while a significant positive trend was found for Bageshwar (0.77), Nainital (0.79), and Pantnagar (0.82) stations at 5% level in the winter season.

Temporal variation of trends in annual rainfall data series

The MK test statistics computed from annual rainfall data series are reported in Table 4 along with their graphical representation in Fig. 6e. Figure 6e illustrates the calculated Z_{MK} values for annual rainfall series and tested at 5% significance level. The significant positive trend was found for Champawat (2.041) station while the significant negative trend was found for Uttarkashi (−2.593) station at 5% level in annual rainfall (Table 4).

The ITA results for all the 13 stations are given in Table 9 and graphically shown in Fig. 11. The significant negative trend was found for Almora (−1.64), Chamoli (−4.49), Dehradun (−3.17), Rudraprayag (−4.92), Tehri Garhwal (−

Table 8 The results of innovative trend analysis on winter rainfall at study stations

Station	Slope (s)	σ	$\rho_{y_1 y_2}$	σ_s	Upper CL	Lower CL
Almora	0.04	69.03	0.986	0.06	0.13	−0.13
Bageshwar	0.77*	50.62	0.976	0.06	0.12	−0.12
Chamoli	−4.05*	96.18	0.974	0.12	0.24	−0.24
Champawat	−0.27*	55.17	0.958	0.09	0.18	−0.18
Dehradun	0.15	61.04	0.969	0.09	0.17	−0.17
Haridwar	0.12	52.30	0.932	0.11	0.21	−0.21
Nainital	0.31*	60.54	0.981	0.07	0.13	−0.13
Pauri Garhwal	0.79*	69.75	0.942	0.13	0.26	−0.26
Pithoragarh	−2.00*	129.19	0.725	0.54	1.06	−1.06
Rudraprayag	−1.01*	74.66	0.986	0.07	0.14	−0.14
Tehri Garhwal	−0.41*	65.44	0.935	0.13	0.26	−0.26
Pantnagar	0.82*	59.51	0.935	0.12	0.24	−0.24
Uttarkashi	−0.19	79.50	0.929	0.17	0.33	−0.33

*Statistically significant at $\alpha = 5\%$ (critical $Z = \pm 1.96$)

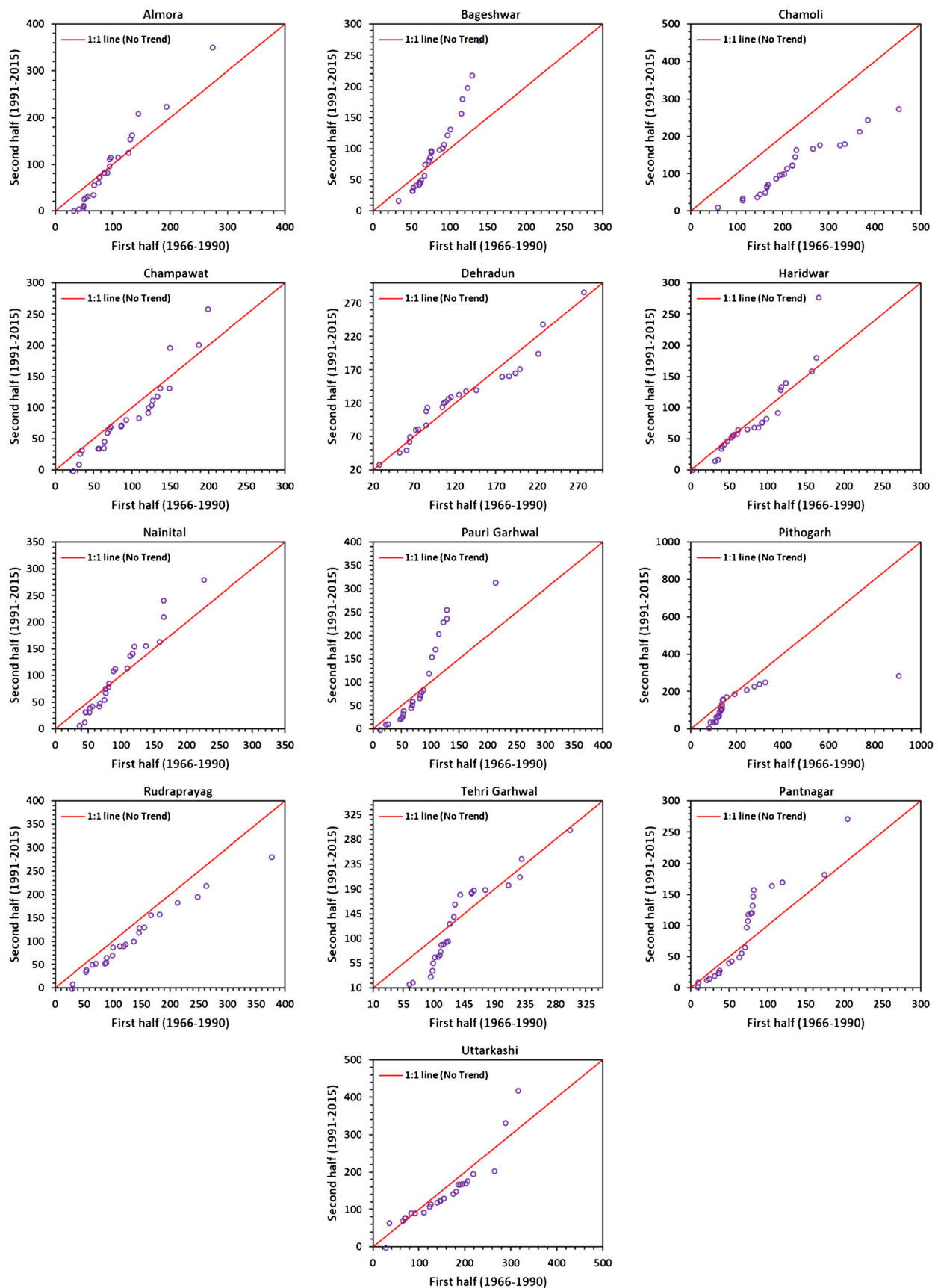


Fig. 10 The results of ITA for winter rainfall at study stations

5.77), and Uttarkashi (− 12.28) stations, while the significant positive trend was found for Champawat (7.70), Haridwar (1.81), Pauri Garhwal (15.61), Pithoragarh (5.75), and Pantnagar (8.63) stations at 5% level in annual rainfall.

Spatial mapping of trend in seasonal and annual rainfall data series

The spatial distribution of seasonal (pre-monsoon, monsoon, post-monsoon, and winter) and annual rainfall trends were identified using the MK and ITA methods. The calculated values of the MK test statistic (Z_{MK}) and ITA (slope values) were spatially mapped for each station with areal coverage using Thiessen polygons developed with the ArcGIS 10.2.2 software. Figures 12 and 13 show the increasing or decreasing spatial trends in seasonal and annual rainfalls at 5% level of significance as represented by various colors viz., (a) ginger pink color for significant negative trends at 5% level; (b) solar yellow color for significantly positive trend at 5% level; and (c) beryl green color for no significant trend at 5% level. Table 3 gives the detailed information about the areas under each station, district, Thiessen polygon, and percent contribution of different districts within each polygon (Fig. 5). Almora polygon with 4046.47 km² (Almora 52.60%, Chamoli 27.97%, Bageshwar 12.41%, Pauri Garhwal 6.79%, and Nainital 0.23%); Bageshwar polygon with 4064.40 km² (Bageshwar 43.73%, Chamoli 32.73%, Pithoragarh 23.43%, and Almora 0.11%); Chamoli polygon with 4654.13 km² (Chamoli 84.88%, Uttarkashi 8.16%, and Pithoragarh 6.95%); Champawat polygon with 3747.18 km² (Champawat 44.9%, Pithoragarh 30.14%, Almora 13.34%, US Nagar 5.87%, Nainital 5.73%, and Bageshwar 0.01%); Dehradun polygon with 3295.78 km² (Dehradun 73.15%, Tehri Garhwal 13.81%, Uttarkashi 11.81%, and Haridwar

1.23%); Haridwar polygon with 3475.18 km² (Haridwar 67.58%, Dehradun 16.41%, Pauri Garhwal 14.07%, and Tehri Garhwal 1.95%); Nainital polygon with 3888.39 km² (Nainital 67.55%, US Nagar 20.92%, and Almora 11.54%); Pauri Garhwal polygon with 4721.67 km² (Pauri Garhwal 90.68%, Nainital 4.38%, US Nagar 3.46%, Tehri Garhwal 1.07%, and Almora 0.41%); Pithoragarh polygon with 4918.90 km² (Pithoragarh 100%); Rudraprayag polygon with 5658.50 km² (Uttarkashi 32.23%, Chamoli 26.75%, Rudraprayag 21.46%, Tehri Garhwal 15.90%, and Pauri Garhwal 3.67%); Tehri Garhwal polygon with 3838.52 km² (Tehri Garhwal 76.02%, Uttarkashi 17.48%, Pauri Garhwal 5.34%, and Dehradun 1.15%); Pantnagar polygon with 2671.40 km² (US Nagar 62.32%, Nainital 37.65%, and Champawat 0.03%); and Uttarkashi polygon with 4831.96 km² (Uttarkashi 99.05% and Dehradun 0.95%).

Spatial mapping of seasonal and annual trends identified by MK and ITA methods

The trends identified by MK test in seasonal and annual rainfall at 5% significance level are spatially mapped in Fig. 12 (Table 4) which reveal that in pre-monsoon season, a significant negative trend was found at 5% level of significance in Chamoli polygon. In the monsoon season, a significant positive trend was detected in Champawat polygon while a negative trend was found in Tehri Garhwal and Uttarkashi polygons at 5% significance level. In the post-monsoon season, a significant negative trend was found in Tehri Garhwal polygon. In the winter season, a significant positive trend was found in the Chamoli polygon while Champawat polygon indicated a negative trend at 5% level of significance. In the annual series, a significant positive trend was found at 5%

Table 9 The results of innovative trend analysis on annual rainfall at study stations

Station	Slope (s)	σ	$\rho_{\bar{y}_1 \bar{y}_2}$	σ_s	Upper CL	Lower CL
Almora	− 1.64*	247.59	0.982	0.26	0.52	− 0.52
Bageshwar	0.40	367.57	0.962	0.57	1.12	− 1.12
Chamoli	− 4.49*	296.12	0.972	0.40	0.78	− 0.78
Champawat	7.70*	515.84	0.938	1.03	2.02	− 2.02
Dehradun	− 3.17*	381.36	0.964	0.58	1.13	− 1.13
Haridwar	1.81*	327.65	0.959	0.53	1.04	− 1.04
Nainital	− 1.37	391.51	0.943	0.75	1.46	− 1.46
Pauri Garhwal	15.61*	568.17	0.917	1.31	2.57	− 2.57
Pithoragarh	5.75*	752.13	0.942	1.45	2.85	− 2.85
Rudraprayag	− 4.92*	417.25	0.932	0.87	1.71	− 1.71
Tehri Garhwal	− 5.77*	347.21	0.941	0.67	1.32	− 1.32
Pantnagar	8.63*	542.52	0.951	0.96	1.88	− 1.88
Uttarkashi	− 12.28*	454.15	0.930	0.96	1.89	− 1.89

*Statistically significant at $\alpha = 5\%$ (critical $Z = \pm 1.96$)

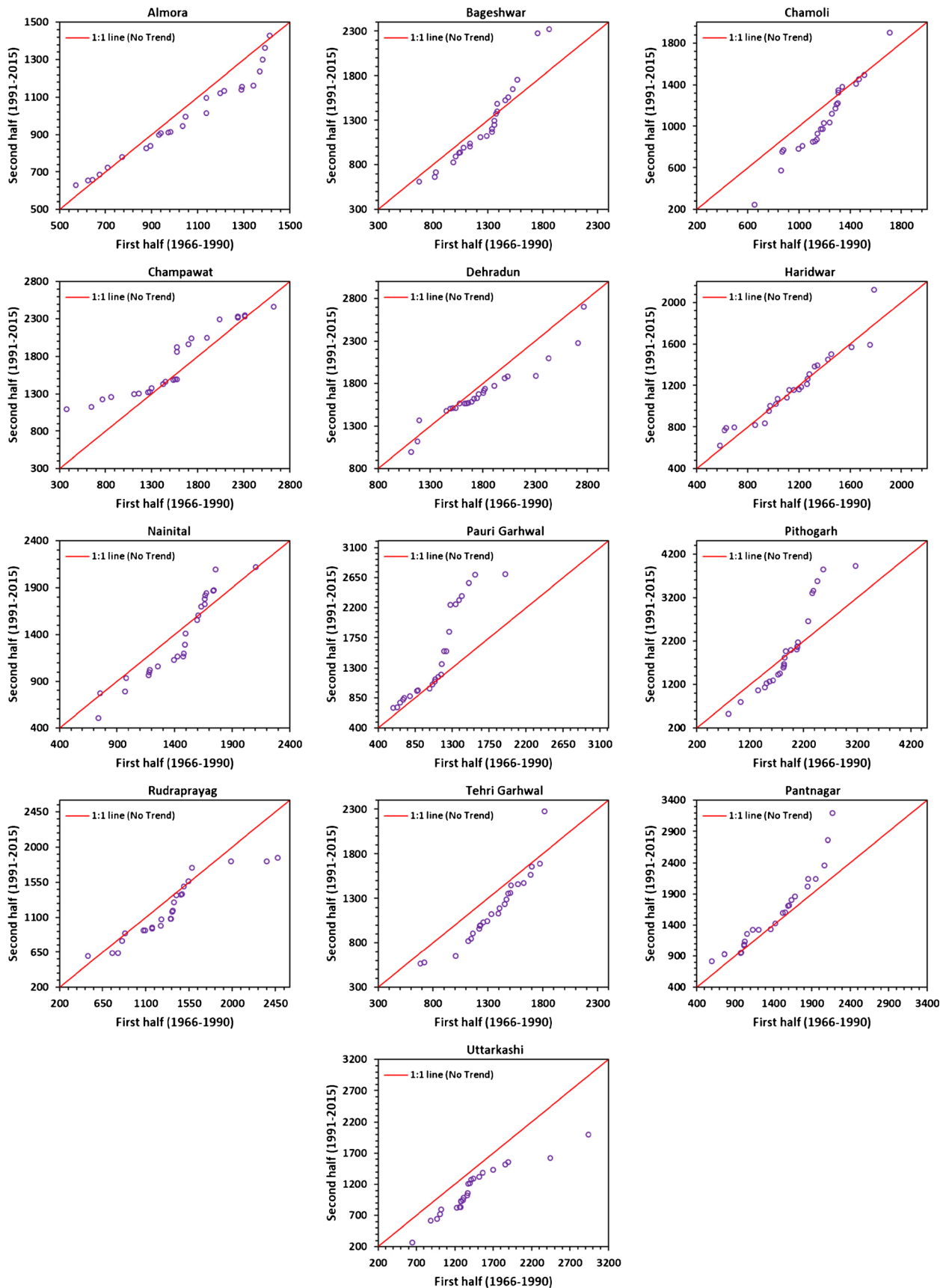


Fig. 11 The results of ITA for annual rainfall at study stations

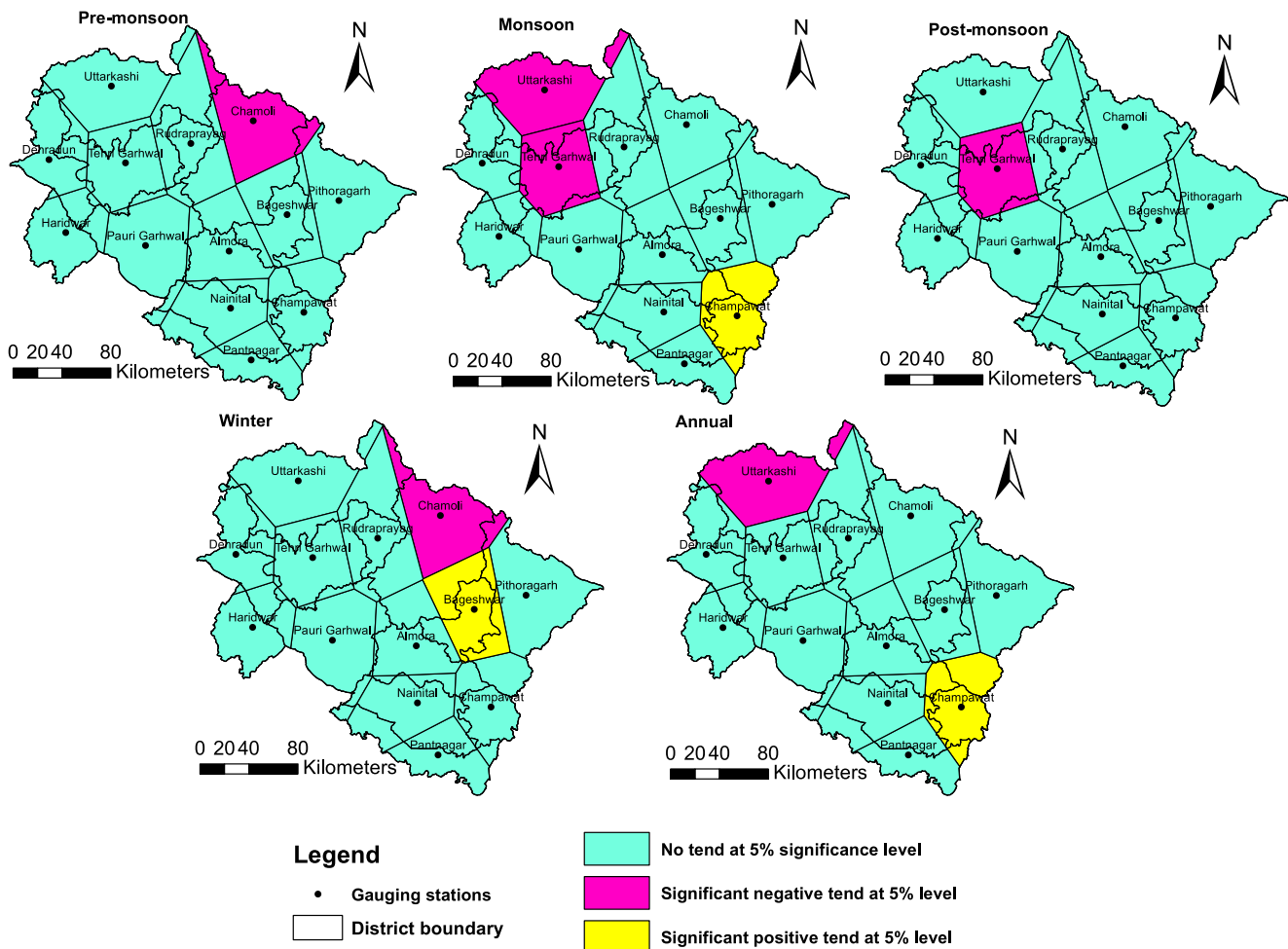


Fig. 12 The spatial variation of seasonal and annual rainfall trends identified by the MK test at 5% significance level

level of significance in the Champawat polygon, while a significant negative trend was found in Uttarkashi polygon.

The trends identified by the ITA method in seasonal and annual rainfall at 5% significance level are spatially mapped in Fig. 13 (Table 5) which reveal that in the pre-monsoon season, a significant negative trend was found in Bageshwar, Chamoli, Champawat, Dehradun, Nainital, and Uttarkashi polygons while Haridwar, Pithoragarh, and Pantnagar polygons showed positive trend at 5% level of significance. For monsoon season, a significant negative trend was found in Almora, Dehradun, Rudraprayag, Tehri Garhwal, and Uttarkashi polygons and a positive trend was found in Chamoli, Champawat, Haridwar, Pauri Garhwal, Pithoragarh, and Pantnagar polygons (Table 6). For the post-monsoon season, a significant negative trend was found in Almora, Bageshwar, Chamoli, Dehradun, Haridwar, Nainital, Pithoragarh, Tehri Garhwal, and Uttarkashi polygons while Pauri Garhwal polygon indicated a positive trend (Table 7). For the winter season, a significant negative trend was found in Chamoli, Champawat, Pithoragarh, Rudraprayag, and Tehri Garhwal polygons and a positive trend was found in Bageshwar, Nainital, and Pantnagar polygons

(Table 8). For annual series, a significant negative trend was found in Almora, Chamoli, Dehradun, Rudraprayag, Tehri Garhwal, and Uttarkashi polygons and a positive trend was found in Champawat, Haridwar, Pauri Garhwal, Pithoragarh, and Pantnagar polygons (Table 9).

Overall, according to the MK test, a significantly decreasing (increasing) trend was observed in 0 (1), 2 (1), 1 (0), 1 (1), and 1 (1) polygons in pre-monsoon, monsoon, post-monsoon, winter, and annual rainfall data, while ITA detected corresponding trends in 7 (3), 5 (6), 9 (1), 5 (4), and 6 (5) polygons in the study region.

Discussion and comparison of trend analysis methods

The above analysis exposed that there were significant increasing and decreasing trends in seasonal and annual rainfall data during 1966–2015 at 13 study stations located in the Uttarakhand State. Meanwhile, there were significant increasing (decreasing) in seasonal (pre-monsoon, monsoon; post-monsoon, winter) and annual rainfall from 1966 to 2015 in Garhwal region (Chamoli, Dehradun, Haridwar, Pauri Garhwal, Rudraprayag, Tehri Garhwal, Uttarkashi) and

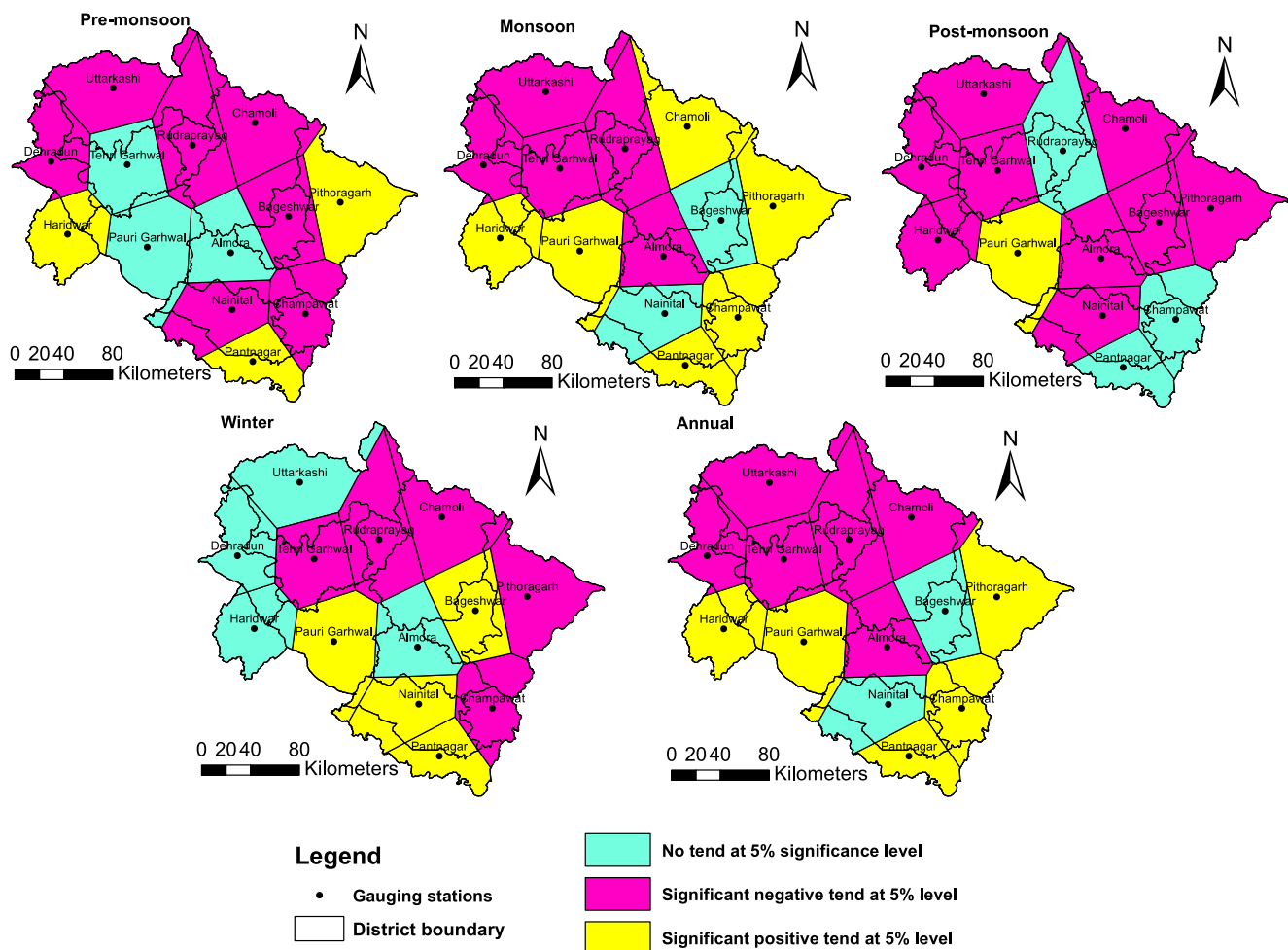


Fig. 13 The spatial variation of seasonal and annual rainfall trends identified by the ITA method at 5% significance level

Table 10 Comparison of results of MK and ITA methods at study stations

Station	Pre-monsoon		Monsoon		Post-monsoon		Winter		Annual	
	MK	ITA	MK	ITA	MK	ITA	MK	ITA	MK	ITA
Almora	No	No	No	Yes (–)	No	Yes (–)	No	No	No	Yes (–)
Bageshwar	No	Yes (–)	No	No	No	Yes (–)	Yes (+)	Yes (+)	No	No
Chamoli	Yes (–)	Yes (–)	No	Yes (+)	No	Yes (–)	Yes (–)	Yes (–)	No	Yes (–)
Champawat	No	Yes (–)	Yes (+)	Yes (+)	No	No	No	Yes (–)	Yes (+)	Yes (+)
Dehradun	No	Yes (–)	No	Yes (–)	No	Yes (–)	No	No	No	Yes (–)
Haridwar	No	Yes (+)	No	Yes (+)	No	Yes (–)	No	No	No	Yes (+)
Nainital	No	Yes (–)	No	No	No	Yes (–)	No	Yes (+)	No	Yes (–)
Pauri Garhwal	No	No	No	Yes (+)	No	Yes (+)	No	Yes (+)	No	Yes (+)
Pithoragarh	No	Yes (+)	No	Yes (+)	No	Yes (–)	No	Yes (–)	No	Yes (+)
Rudraprayag	No	No	No	Yes (–)	No	No	No	Yes (–)	No	Yes (–)
Tehri Garhwal	No	No	Yes (–)	Yes (–)	Yes (–)	Yes (–)	No	Yes (–)	No	Yes (–)
Pantnagar	No	Yes (+)	No	Yes (+)	No	No	No	Yes (+)	No	Yes (+)
Uttarkashi	No	Yes (–)	Yes (–)	Yes (–)	No	Yes (–)	No	No	Yes (–)	Yes (–)

(±) and No indicate significant increasing (decreasing) and no trends at 5% significant level

Kumaon region (Almora, Bageshwar, Champawat, Nainital, Pithoragarh and US Nagar/Pantnagar). In the past years, throughout the world, a number of studies have been conducted on spatio-temporal trend analysis of meteorological data using parametric and non-parametric methods such as (Brunetti et al. 2012; Ay and Kisi 2015; Caloiero et al. 2017; Mohorji et al. 2017; Caloiero et al. 2018; Güçlü 2018; Balasmeh et al. 2019). The spatial variability of rainfall trend is very important for impact assessment and adaptation of planning for floods, droughts, and extreme events (Terêncio et al. 2017, 2018) in the study region.

In this study, a total of 65 series of seasonal and annual rainfall were analyzed using MK and ITA methods at 5% significance level. The results of the comparison of MK and ITA methods are summarized in Table 10. It was observed from the table that the significant (positive and negative) trends were detected only in 9 series by the MK, while the ITA detected significant (positive and negative) trends in 51 series. Moreover, almost all significant trends (at 5% level) that can be detected by the MK test (9-time series) can be effectively detected using ITA (51-time series). The ITA method detected some hidden significant trends which could not be identified by the MK method in seasonal and annual rainfall series. This would be beneficial for analyzing the hidden variation of trends of rainfall series accurately in the study region. The results of the comparison also revealed that the ITA method was superior to the MK method. In the ITA method, the data can be illustrated graphically on the Cartesian coordinate system; therefore, the visual inspection can be accounted for further interpretations (Şen 2017). Furthermore, the ITA method has the following advantages, such as (i) it is simple to understand and calculate and (ii) it does not depend on any kind of distribution assumption, serial correlation, and seasonal cycle (Cui et al. 2017; Şen 2017; Zhou et al. 2018). This study also confirmed the supremacy of the ITA method over the MK test in detecting the hidden trend in seasonal and annual rainfall time series data for the study region.

Conclusions

In the present study, the trends in seasonal (pre-monsoon, monsoon, post-monsoon, and winter) and annual rainfall time series were detected for 13 stations of the Uttarakhand State, India, using Mann-Kendall and innovative trend analysis methods at 5% significance level. The spatial pattern of trends (increasing and decreasing) was interpolated using Thiessen polygon method in ArcGIS 10.2 environment. The following specific conclusions are derived from this study:

- The results of the MK method revealed that there were significant positive and negative trends in the 9-time series of seasonal and annual rainfall.

- The results of the ITA method revealed that there were significant positive and negative trends in 51-time series of seasonal and annual rainfall.
- Almost all the significant trends (5% level) detected using the MK method can be effectively identified by the ITA method, which indicates that the ITA method is more sensitive to trends than the MK method.
- The spatial pattern of trends detected using MK and ITA methods at 5% significance level was interpolated using Thiessen polygon method in ArcGIS 10.2 environment.
- The developed spatial maps would help local stakeholders and water managers to understand the risks and vulnerabilities related to climate change due to the anthropogenic factors in the region.

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